Hybrid Family: Loss & Overlap Breakdown

shg_mode_survey · hybrid_sweep1 (lateral) + hybrid_vertical1 (vertical)

376 refined pump→SH crossings across 47 geometries -- AlN core embedded IN an SiNx shell (not stacked on top; see the 'pedestal' report for that earlier, accidental structure). First hybrid-family campaign run with the 2026-10-12 SH-loss calibration wired in from launch (100 dB/cm flat, confinement-weighted, SH-band only).

SH = 215-235 nm, deep UV. Pump = 430-470 nm, true blue.

WHAT THIS CAMPAIGN WAS BUILT TO TEST

1. Does the Honda et al. TM00->TM20 mechanism (or a close analogue) actually appear now that AlN is correctly embedded? NOT exactly -- no crossing hits TM20 itself. Closest analogues are higher-vertical-order TM2n (n=5-9) crossings; see the dedicated page below.
2. Does the pedestal data's 'smaller sin_height keeps winning' trend continue to the new sin_height=450 boundary point? NO -- it reverses. Best NCE peaks at sin_height=500, not at the smallest tested value; see the dedicated page below.

Both are genuine, useful answers -- a campaign that confirmed the naive hypothesis either way would have told us less than finding the real optimum sits in the interior of the range.

CALIBRATED SH LOSS / TOTAL SH ABSORPTION
0.007% median
native SiNx shell bulk UV absorption swamps the calibrated device-anchored term

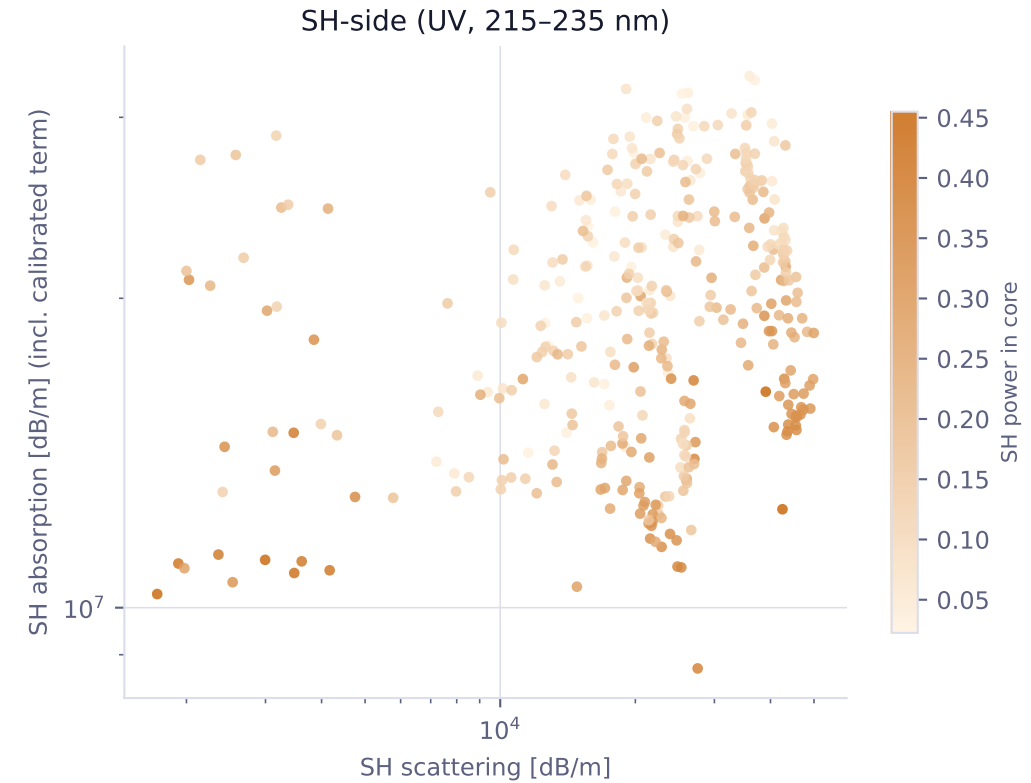
SH ABSORPTION / SCATTERING
832× median
same dominance pattern as the pedestal family, even stronger here

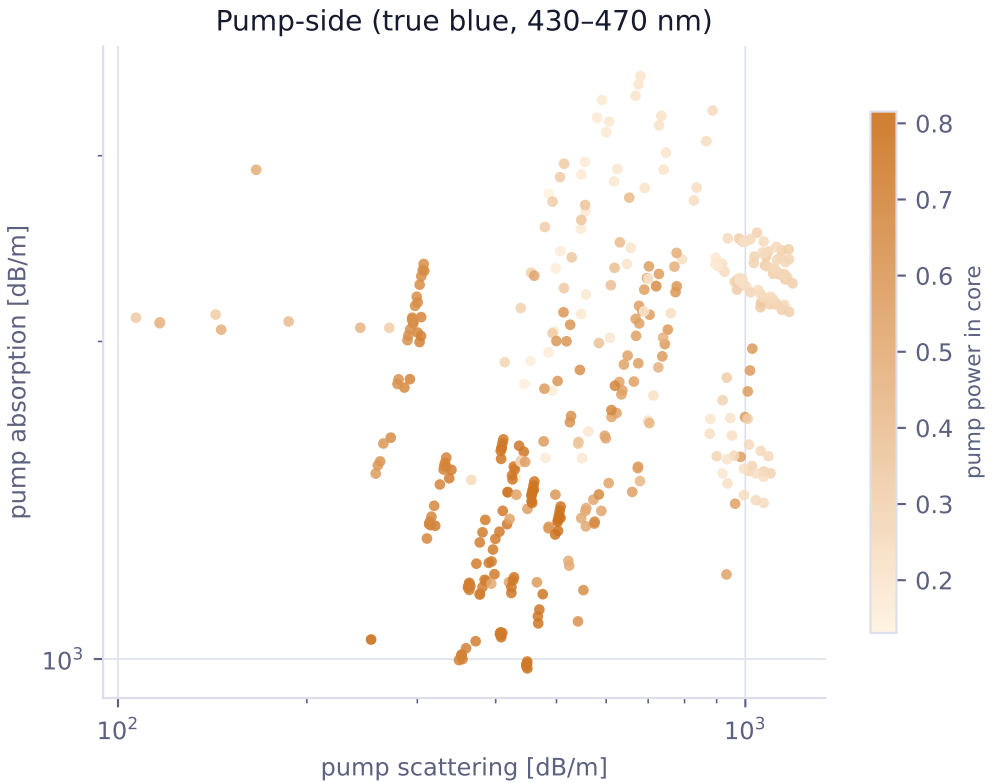
SH LOSS / PUMP LOSS
7979× median
same crossing, UV vs. blue -- very different bands

OVERLAP_SHAPE ↔ LOG(ETA)
r = 0.44
shape overlap alone barely predicts final efficiency

Best crossing overall: TM10→TM100 @ 223.80 nm, eta=125.8 %/W/cm² (h=250/w=400/sin_w=700/sin_h=500, run=hybrid_vertical1)

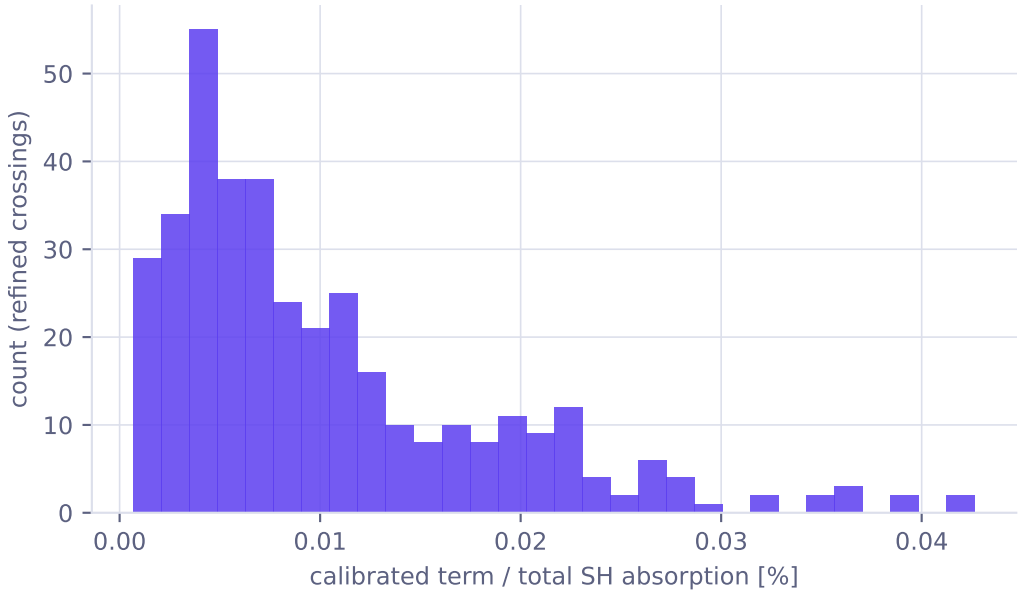
Loss breakdown: scattering vs. absorption

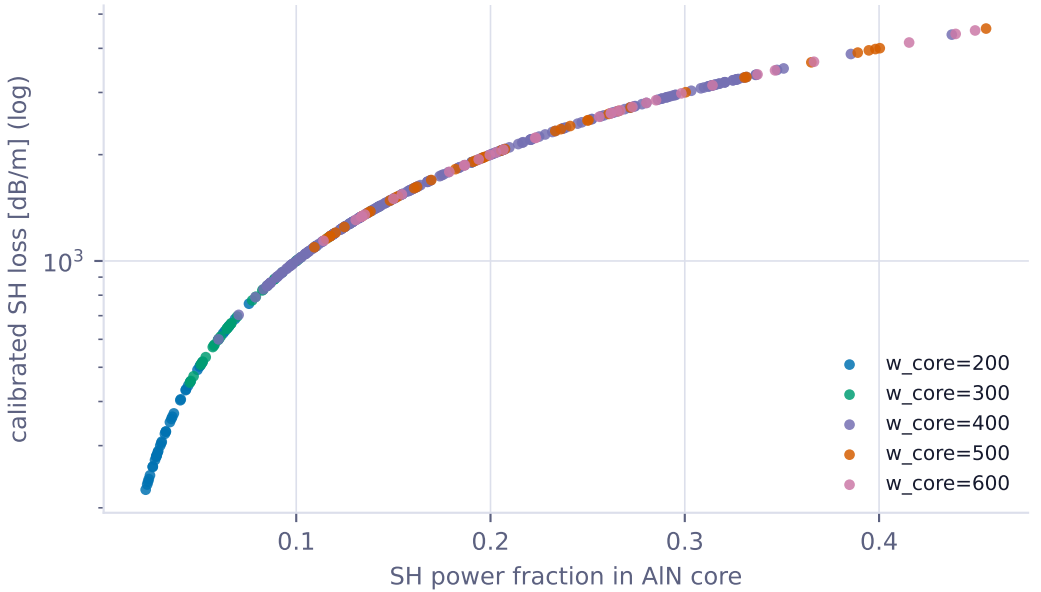




SH absorption on the left already includes the calibrated SH-loss term (sh_calibrated_dB_per_m) summed in -- see the next page for how small a share of the total that ter

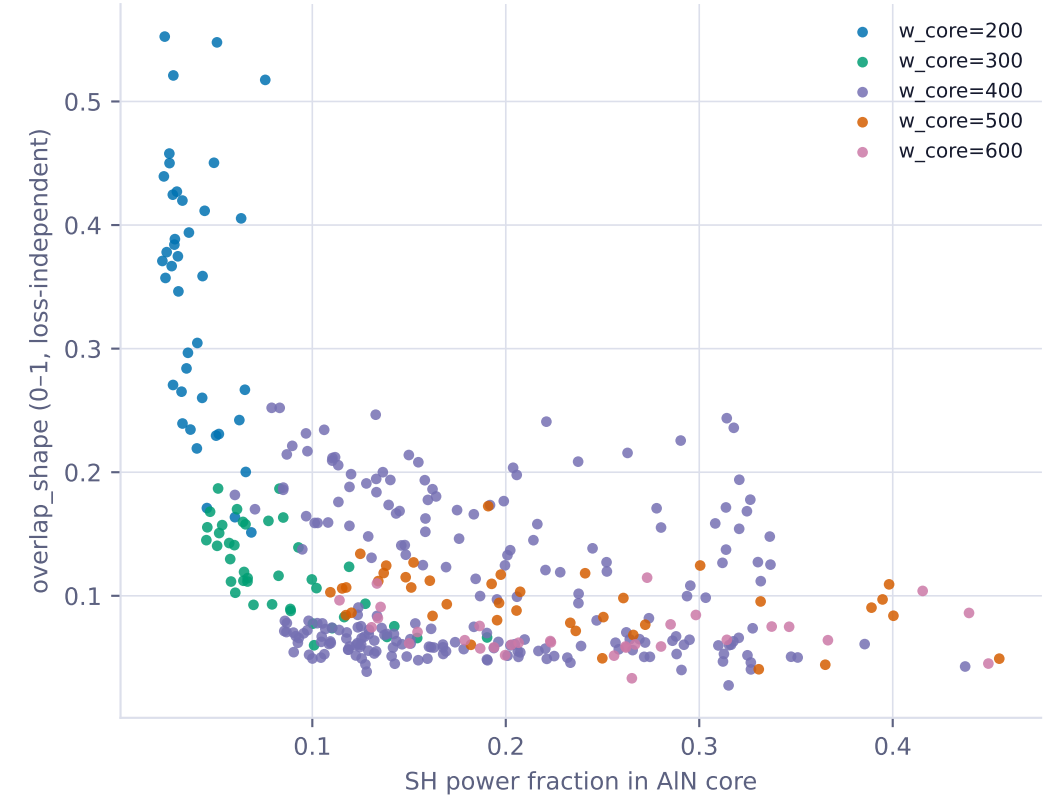
Calibrated SH loss vs. native SiNx shell absorption

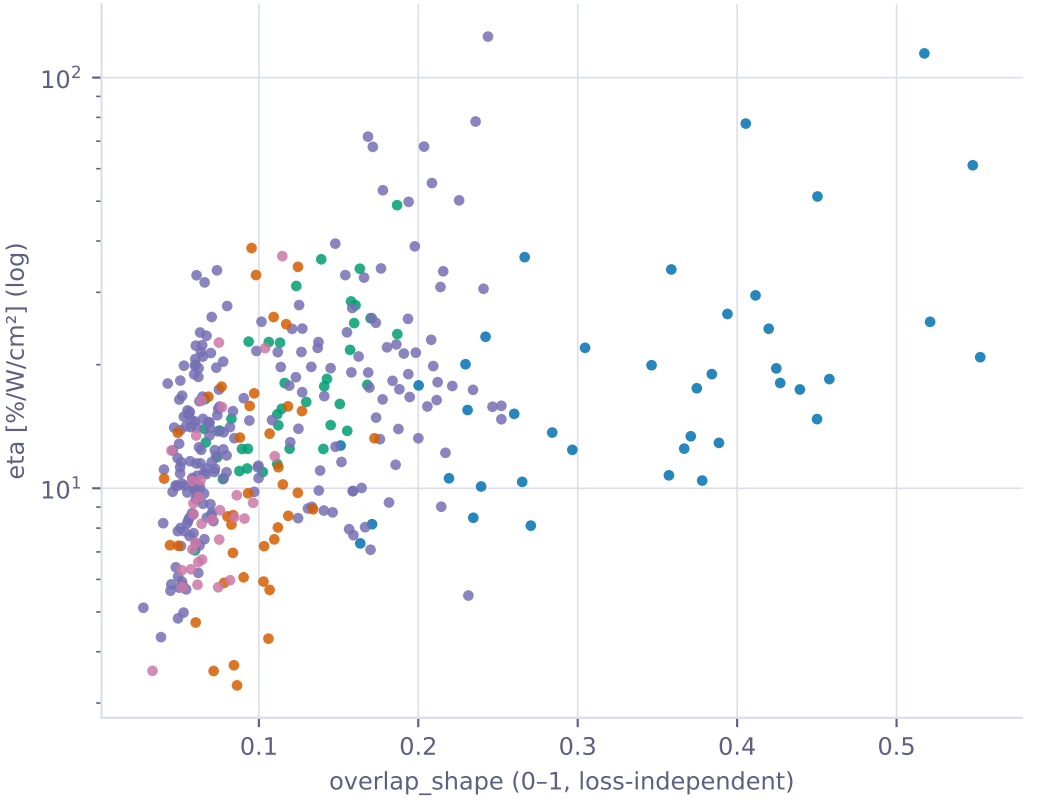




Median calibrated share of total SH absorption: 0.007%. The device-anchored 100 dB/cm estimate (right panel, scales linearly with confinement as designed) is real but 13,400x smaller than the native SiNx shell's own UV bulk absorption at these overlaps -- the AM187 story mattered for setting a sane SH-loss scale, but in this geometry family it's not the dominant term.

Overlap vs. confinement vs. final efficiency



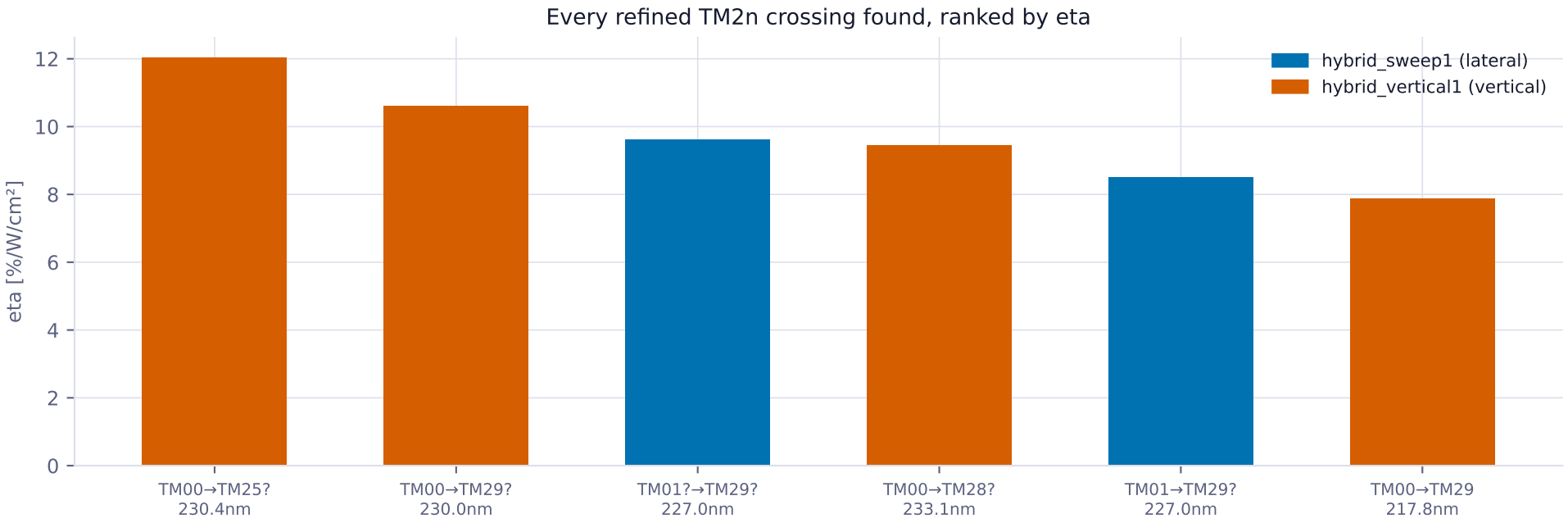


Mechanism search: did TM00→TM20 (or an analogue) appear?

Honda et al. 2025: AlN confined to one sign-lobe of a lateral TM2n mode lets d33 avoid self-cancellation across the mode's alternating-sign lobes -- the mechanism this lateral/vertical sweep pair was purpose-built to find.

RESULT: not found exactly, but close analogues exist

No refined crossing in either run hits SH label TM20 itself. 6 refined crossings hit the broader TM2n family (m=2 lateral, any vertical order) -- all at n=5-9, not n=0. That IS the right lateral order (m=2, the self-cancellation-avoiding signature), just with more vertical structure than the idealized target.



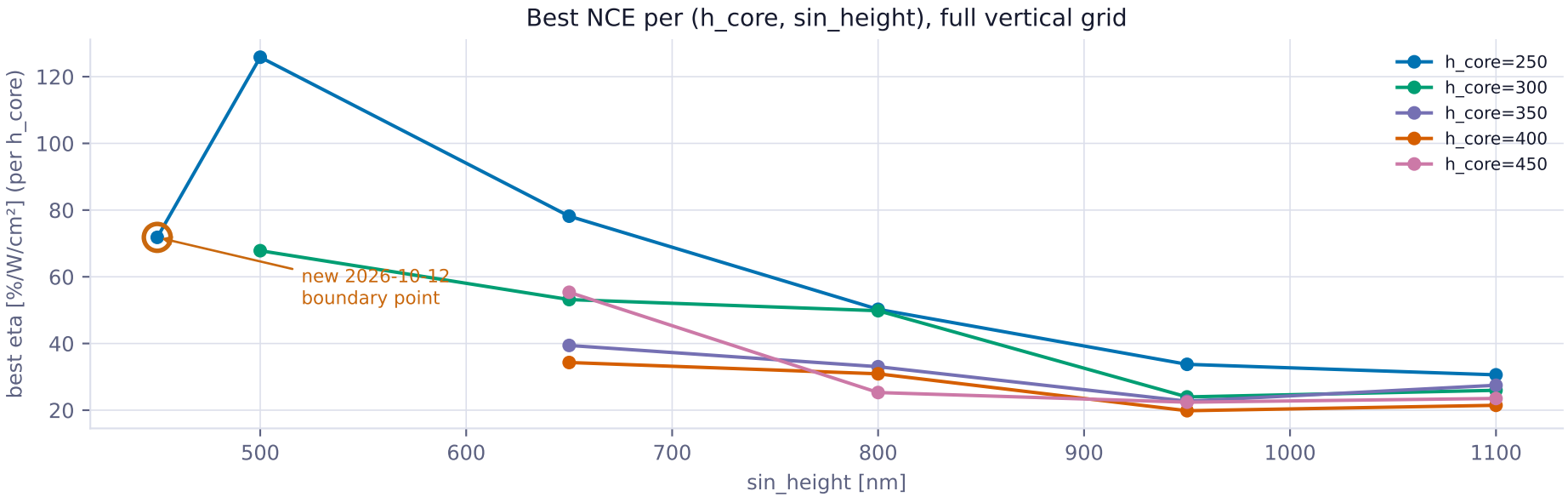
Reading: these ARE the m=2 lateral family the mechanism predicts, achieving competitive NCE (up to ~12 %/W/cm² in the vertical sweep, ~10 %/W/cm² in the lateral sweep) despite not being the literal TM20. Worth a closer look at the actual field plots before concluding whether the extra vertical nodes are incidental or point to a geometry that would reach true TM20 with a different sin_height/sin_width combination than searched here -- not done as part of this report.

Vertical sweep: does smaller sin_height keep winning?

The pedestal family's archival data trended toward smaller sin_height across 500-1100 nm without bottoming out in that range -- motivating this sweep's new sin_height=450 boundary point (h_core=250, the smallest margin-valid combination).

RESULT: no -- the trend reverses, peak is at sin_height=500

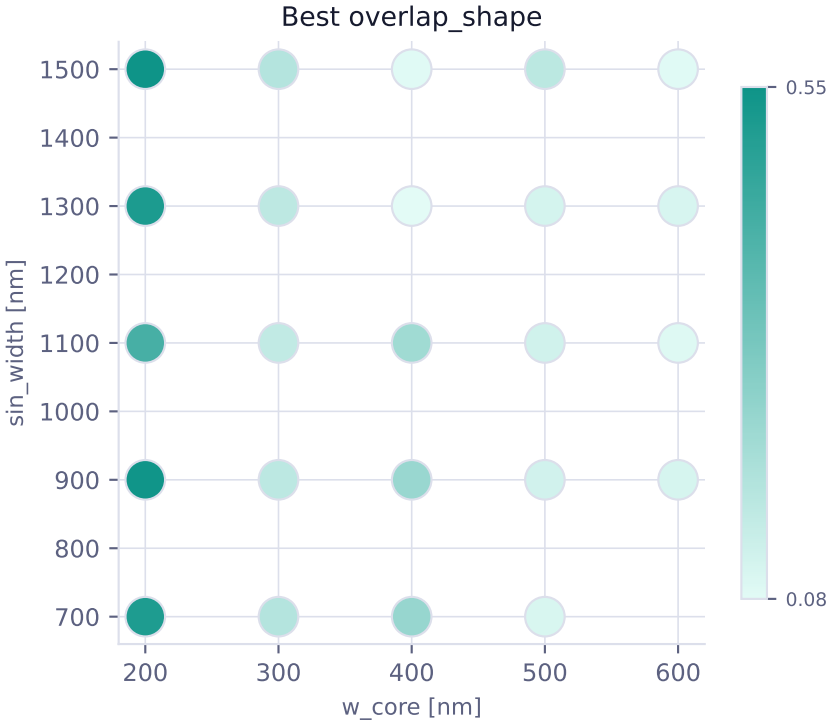
Best NCE at h_core=250: 71.8 %/W/cm² at sin_height=450 vs. 125.8 %/W/cm² at sin_height=500 -- the new boundary point is WORSE than its neighbor, not better. The real optimum sits in the interior of the tested range, not at either edge.

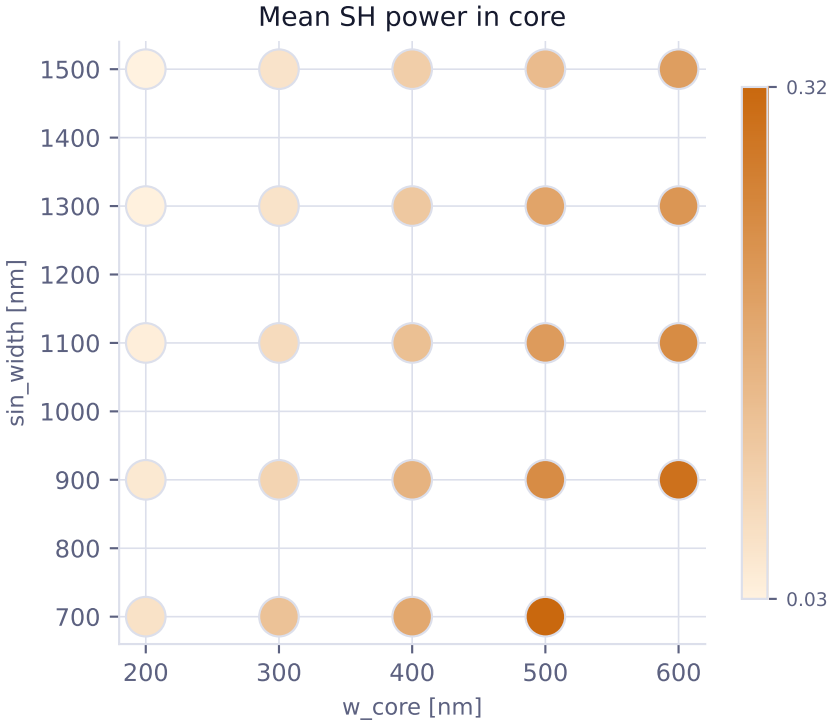


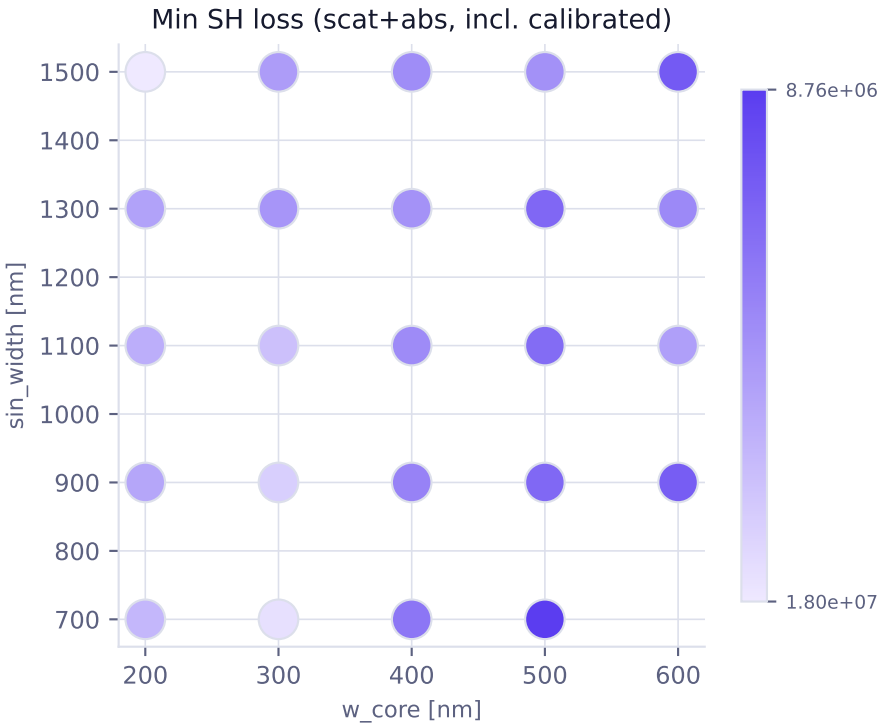
Every h_core row shows the same qualitative shape where tested widely enough: NCE rises from the tall/loose end down to an interior maximum, then falls again toward the tightest margin (more confinement helps up to a point, then the mode distorts/loss rises faster than overlap gains). h_core=250 is the only row dense enough near the boundary to resolve the turnaround explicitly; the global best (TM10→TM100, 125.8 %/W/cm²) sits right at that interior peak.

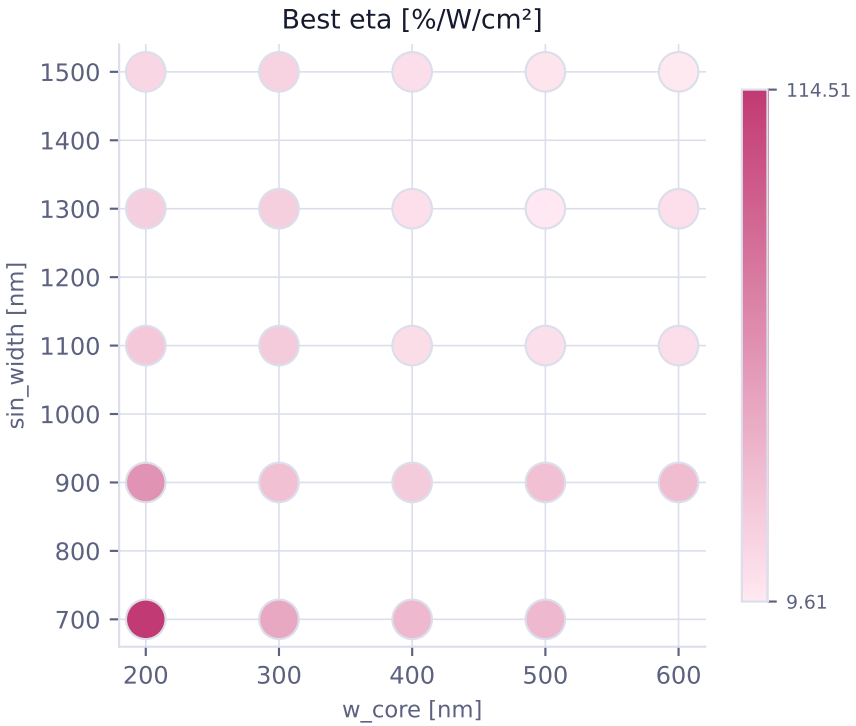
Lateral geometry maps

Fixed h_core = 400 nm, sin_height = 700 nm



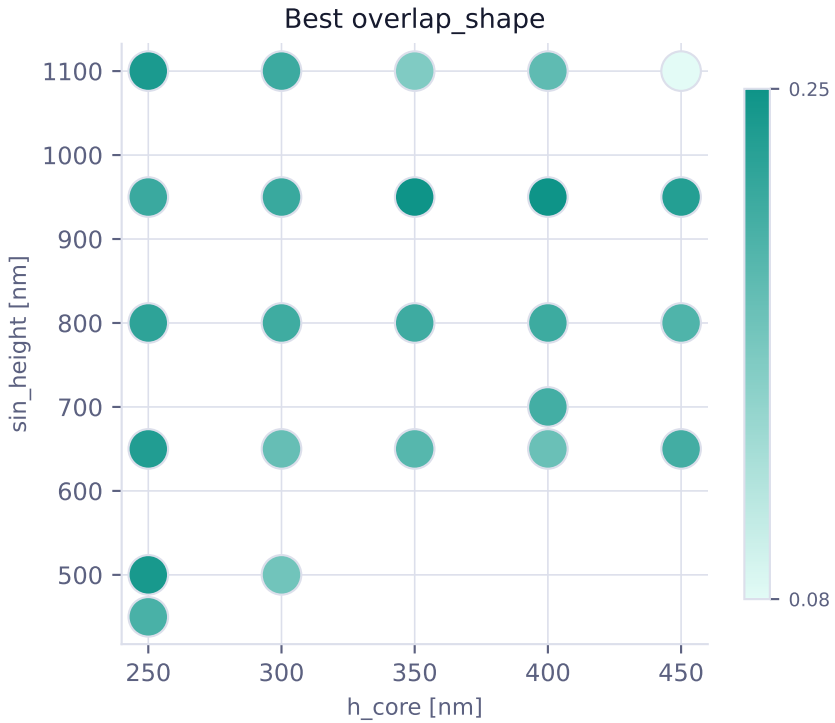


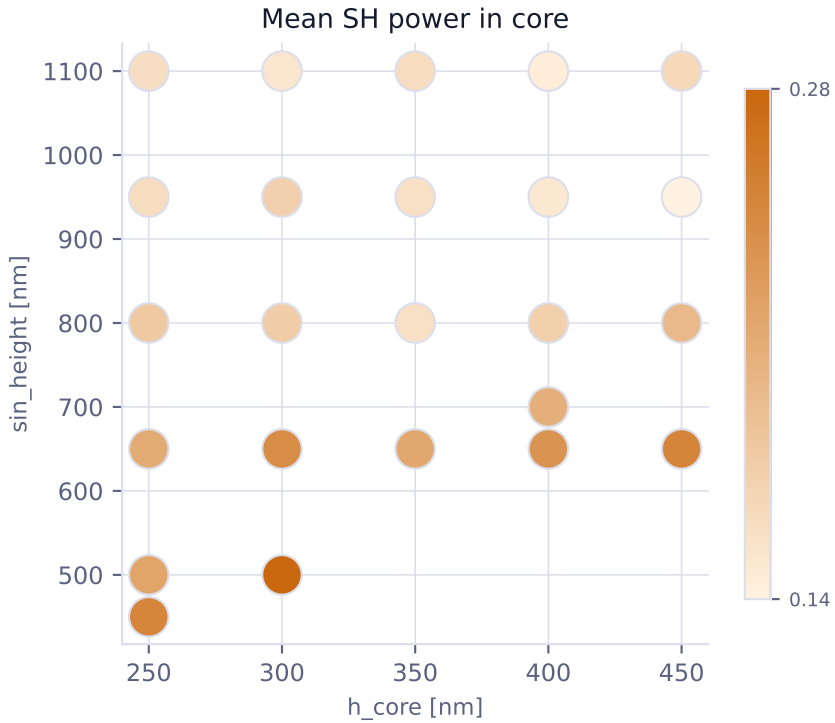


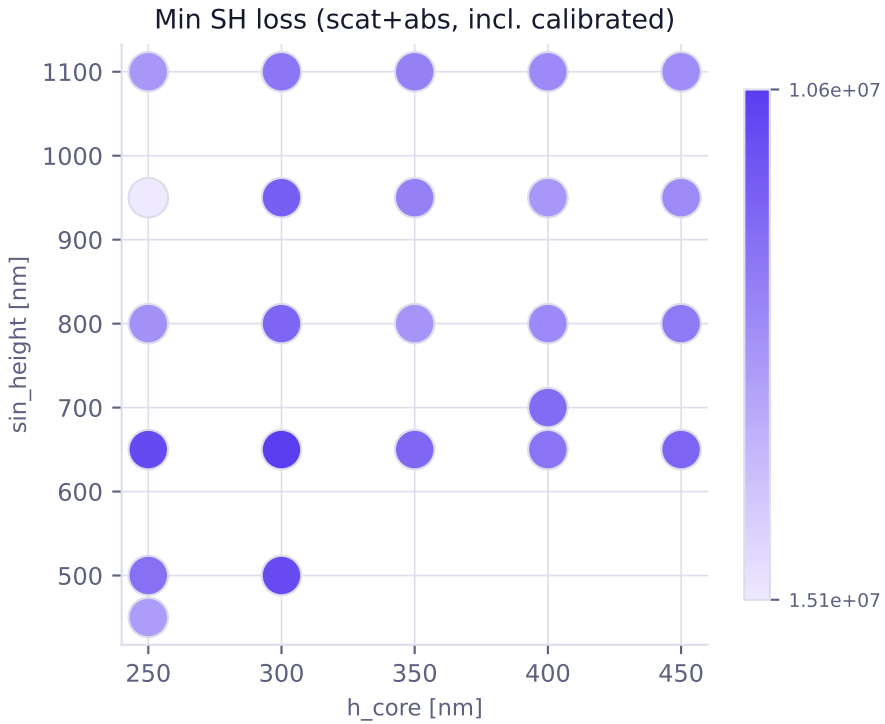


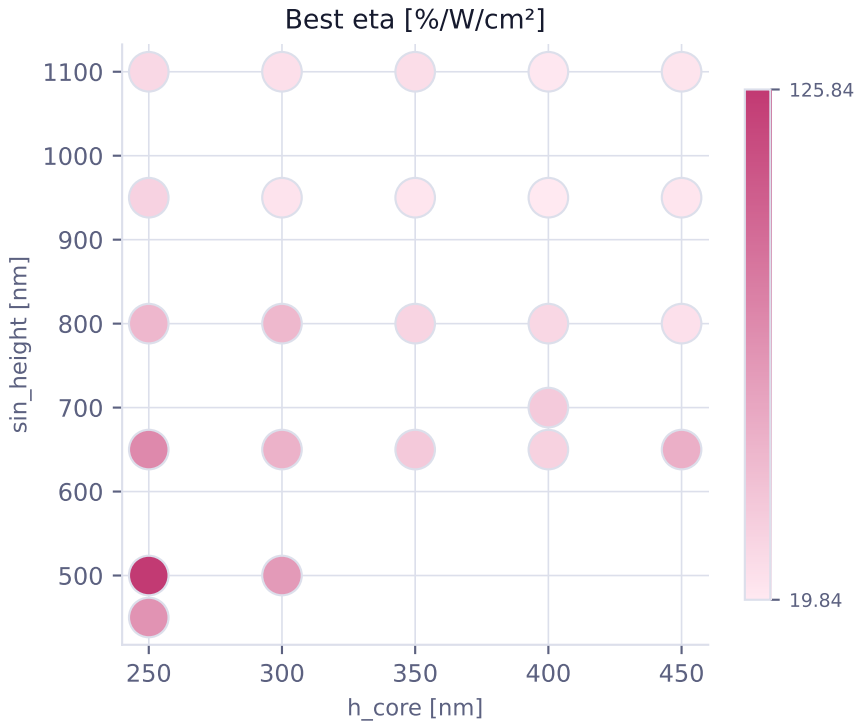
Vertical geometry maps

Fixed w_core = 400 nm, sin_width = 700 nm

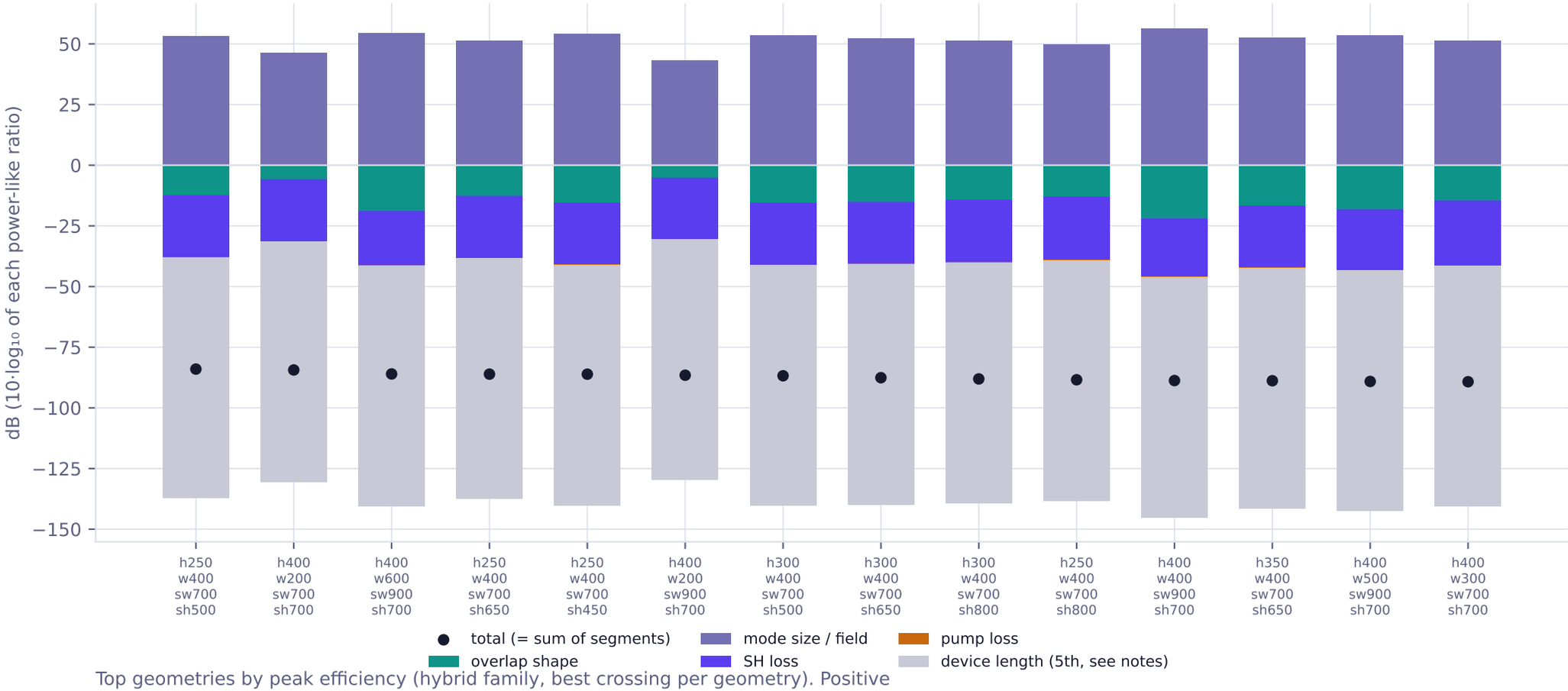








NCE breakdown: dB-additive waterfall per geometry



Top geometries by peak efficiency (hybrid family, best crossing per geometry). Positive segments stack upward, negative downward, from 0 -- the black dot always lands exactly on their sum. 'Mode size / field' is the one unbounded segment (a physical scale set by confinement and the d-tensor, not a 0-1 quality factor like the other three); everything else is <= 0 dB by construction (overlap_shape <= 1, loss always reduces L_eff).

'Device length' is a 5th, unrequested-but-necessary term: each bar is evaluated at ITS OWN optimal propagation length (loss_limited_length's L_opt, same quantity behind this project's usual peak_efficiency_pct_per_W) rather than one shared length -- without it the other four terms alone would not sum to the true total. See nce_decomposition.py's docstring for the two earlier (rejected) versions that didn't reconstruct the total on real data.

Notes & provenance

Data: runs/hybrid_sweep1 (lateral, w_core x sin_width @ fixed h_core=400/sin_height=700) and runs/hybrid_vertical1 (vertical, h_core x sin_height @ fixed w_core=400/sin_width=700) crossings.csv, refined rows only. Extracted via extract_subfigure_data.py, rendered via build_hybrid_subfigure_report.py. Both launched together 2026-10-12 overnight, completed clean the next morning: 24/24 and 23/23 geometries 'ok', zero errors, cerenkov_allowed_any False on every refined crossing in both runs (nothing to flag per the standing checklist item).

This is the CORRECTED 'hybrid' family -- AlN core embedded inside an SiNx shell, the actual Honda et al. 2025 structure -- not the 'pedestal' family (AlN stacked on top of SiNx), which is what every hybrid-named run before the 2026-10-10 geometry-construction fix had actually been building by accident. See the pedestal report and PLAN.md for that earlier story.

sh_absorption_dB_per_m here already has the 2026-10-12 SH-loss calibration term summed in at refine time (survey.py's refine_crossing: abs_s = mechanisms + bulk_s + cal_s); the calibrated contribution alone is separately available as sh_calibrated_dB_per_m, used on the calibrated-fraction page above. Do not re-add it via nce_decomposition.decompose(..., include_sh_calibrated=True) on these rows -- that flag is for reprocessing OLDER rows that don't have it baked in; calling it here double-counts.

overlap_shape is dimensionless 0-1, integrated over the AlN region only -- 1 means the SH field inside AlN is exactly proportional to the pump-driven polarization; it carries no information about HOW MUCH SH power actually reaches AlN (that's sh_power_in_core, measured separately).

AlN is still modeled as lossless in its own right throughout (its own UV k remains a separate, open item, deliberately rejected as the SH-loss basis 2026-10-12 -- see build_assumptions_report section 3b); every absorption number above is SiNx shell bulk absorption plus the calibrated device-anchored term, never AlN's own intrinsic loss.